

JANUARY 11, 2018

New 'emotional' robots aim to read human feelings



Buddy the companion robot by Blue Frog Robotics is seen on display during the CES Unveiled preview event at the Mandalay Bay Convention Center during CES 2018 in Las Vegas on Jan 7, 2018. Credit: AFP/Mandel Ngan

The robot called Forpheus does more than play a mean game of table tennis. It can read body language to gauge its opponent's ability and offer advice and encouragement. "It will try to understand your mood and your playing ability and predict a bit about your next shot," said Keith Kersten of Japan-based Omron Automation, which developed Forpheus to showcase its technology. "We don't sell ping pong robots but we are using Forpheus to show how technology works with people," said Kersten.

Forpheus is among several devices shown at this week's Consumer Electronics Show which highlight how robots can become more humanlike by acquiring "emotional intelligence" and empathy.

Although this specialization is still emerging, the notion of robotic empathy appeared to be a strong theme at the huge gathering of technology professionals in Las Vegas.

Honda, the Japanese auto giant, launched a new robotics program called Empower, Experience, Empathy including its new 3E-A18 robot which "shows compassion to humans with a variety of facial expressions," according to a statement.

Although empathy and emotional intelligence do not necessarily require a humanoid form, some robot makers have been working on form as well as function. "We're been working very hard to have an emotional robot," said Jean-Michel Mourier of French-based Blue Frog Robotics, which makes the companion and social robot called Buddy, set to be released later this year. "He has a complex brain," Mourier said at a CES event. "It will ask for a

caress or it will get mad if you poke him in the eye." Other robots such as Qihan Technology's Sanbot and SoftBank Robotics' Pepper, are being "humanized" by teaching them to read and react to people's emotional states.



The Omron Forpheus robot plays table tennis against a human at CES in Las Vegas. (AFP)

Pepper is "capable of interpreting a smile, a frown, your tone of voice, as well as the lexical field you use and non-verbal language such as the angle of your head," according to SoftBank.

Robot in human shoes

Developing emotional intelligence in robots is a difficult task, melding the use of computer "vision" to interpret objects and people and creating software that can respond accordingly.

"Empathy is the goal: the robot is putting itself in the shoes of the human, and that's about as hard as it gets," said Patrick Moorhead, a technology analyst with Moor Insights & Strategy.

"It's not just about technology, it's about psychology and trust." Moorhead said this technology is still in the early stages but holds promise in some areas, noting that there is strong interest in Japan amid a lack of caretakers for the elderly population. "In some ways it can be a bit creepy if you're crying and the robot is trying to console you," he said. "If you have no friends, the next best thing is a friend robot, and introverts might feel more comfortable talking to a robot."

'Emotion chip'

One CES exhibitor offered a promise of going further than the current devices by developing an "emotion chip" which can allow robots to process emotions in a manner similar to humans.

"There has been a lot of research on detecting human emotions. We do the opposite. We synthesize emotions for the machine," said Patrick Levy-Rosenthal, founder of New York-based Emoshape, which is producing its chip for partners in gaming, virtual and augmented reality and other sectors. It could be used to power a humanoid robot, or other devices. For example, an e-reader could better understand a text to infuse more emotion in storytelling.

As for Forpheus, Kersten said the robot's ability to help people improve their table tennis skills could have numerous applications for sports, businesses and more. "You could sense how people are feeling, if they are attentive or in a good state to drive," he said.

Another key application could be in health care, he said: "In an elderly patient facility, you can determine if someone is in distress and needs help."

<https://techxplore.com/>

Lea el artículo y responda las siguientes consignas:

1. Explique las características de los robots mencionados.
2. ¿Qué quiere decir el autor con el subtítulo “Robots in human shoes”?
3. En el cuarto párrafo se encuentra el conector **ALTHOUGH** marcado en negrita. ¿Qué tipo de conector es? ¿Cuáles son los dos conceptos que relaciona?
4. Sintetice el artículo en no más de cuatro oraciones, teniendo en cuenta los conceptos principales.

TEXTO 26 – EXPERIMENT

Pre lectura

1. Lea solo los títulos. ¿Qué tipo de texto es? ¿En qué asignatura lee este tipo de texto?
2. ¿De qué trata?
3. Organización textual. ¿Cuáles son las partes a seguir en un experimento?
4. Compare esta guía con las guías de laboratorio de UTN. ¿Sigue el mismo orden?

Experiment 10

Acetic Acid Content of Vinegar: An Acid-Base Titration

The task

The goal of this experiment is to determine accurately the concentration of acetic acid in vinegar *via* volumetric analysis, making use of the reaction of acetic acid with a strong base, sodium hydroxide.

Skills

At the end of the laboratory session you should be able to:

- use an analytical balance,
- use a pipette filler and a pipette,
- use a volumetric flask to make up a solution of a given concentration accurately,
- use a burette to carry out a titration.

Other outcomes

You will develop an understanding of how an acid-base indicator can be used to establish the end-point of a titration.

The Assessment

You will be assessed on your ability to use the pipette and pipette filler. See Skill 4.3.

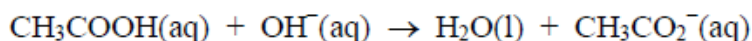
Introduction

Vinegar is an acidic liquid, which is made both naturally and synthetically, from the oxidation of ethanol, $\text{CH}_3\text{CH}_2\text{OH}$, in an alcohol-containing liquid such as wine, fermented fruit juice (*e.g.* cider) or beer. It has been used since ancient times as an important cooking ingredient, *e.g.* in salad dressings and on fish and chips. The key chemical component of vinegar is acetic acid, CH_3COOH (systematic name: ethanoic acid). The trivial name, acetic acid, is derived directly from the word for vinegar, which, for example, in Italian is called *aceto*. The word vinegar itself derives originally from the Latin *vinum aegrum*, meaning “feeble wine”. In fact, when a wine has “gone off” and has acquired a sour taste, this is due to the oxidation of the ethanol in the wine to acetic acid. (The “corking” of wine, *i.e.* tainting of the wine by compounds transferred from or through the cork, is due to a totally different chemical process.) Vinegar derived from red or white wine is the most commonly used form of vinegar in Mediterranean countries and central Europe. The acetic acid content of vinegar can vary widely, but for table vinegar it typically ranges from 4 to 8 % v/v. When used for pickling, the acetic acid content can be as high as 12 %. The purpose of this experiment is to determine the acetic acid content of a commercial vinegar by volumetric analysis.

Volumetric analysis

Volumetric analysis (see Skill 4) is a technique that employs the measurement of volumes to determine quantitatively the amount of a substance in solution. In any reaction between two or more species, the reaction equation shows the stoichiometric ratio of reacting species. Hence, if the concentration of one of the solutions is known, the concentrations of the others can be determined from the volumes used.

Take, for example, the reaction being investigated today, that between solutions of acetic acid and sodium hydroxide, NaOH. The reaction equation is:



The OH^- (from sodium hydroxide) is present at a known molar concentration. A fixed volume of OH^- solution is taken and the CH_3COOH solution progressively added until the point at which complete reaction of the substances is reached. This is called the equivalence point, *i.e.*, the point at which the moles of CH_3COOH added equals the number of moles of OH^- present in the original solution. The incremental process of adding the CH_3COOH solution is called titration. It enables the concentration of the CH_3COOH to be determined from the stoichiometric equation and the volume of the CH_3COOH solution needed to reach the equivalence point.

pH Indicators

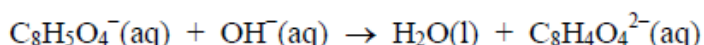
Up to the equivalence point the reacting solution contains an excess of one reactant (here OH^-); after the equivalence point the solution contains an excess of the other reactant (here CH_3COOH). We need some means of observing this switch-over. In acid/base titrations (to which the acetic acid/hydroxide titration belongs), the change in the colour of a pH indicator is often used to identify the switch-over. The point at which this observed change occurs is called the end point of the reaction. An appropriate indicator is one in which the end point and the equivalence point are as close together as possible.

pH indicators themselves are generally both weak acids and strongly coloured dyes, which respond with a dramatic change in colour when they react with base, *i.e.* OH^- . Because they react with base, in principle they compete with the acid whose concentration you are trying to measure (*i.e.* CH_3COOH). However, because the indicators are so highly coloured, very low concentrations are required, and you can generally ignore perturbation of the titration by the indicator in your calculations. However, it is important to keep the indicator concentration low.

In this experiment you will use the indicator phenolphthalein, which undergoes a sudden change in colour from pink in basic solution to colourless in acidic solution. The midpoint of the colour change occurs at a pH of 9.5, *i.e.* already slightly basic. This is a perfect choice as an indicator for the titration of acetic acid with hydroxide ion, because the solution at the equivalence point of this titration is slightly basic.

Standardisation

Solutions of sodium hydroxide react slowly with carbon dioxide gas present in the atmosphere. Consequently, the solution of sodium hydroxide which you will use in today's experiment needs to be standardised (*i.e.* its concentration accurately determined) before use. This is done by titrating with a solution of a stable acidic compound of high purity which can be very accurately prepared to a known concentration. This is known as the primary standard. In this experiment you will use potassium hydrogenphthalate, $\text{KC}_8\text{H}_5\text{O}_4$, for this purpose. The equation for its reaction with hydroxide ion is:



In this experiment, you will perform a series of acid/base titrations. In the course of these titrations you will become familiar with the technique of titration and the calculations associated with volumetric analysis. Read the Skills sections on weighing by difference (Skill 3.2) and on volumetric analysis (Skill 4.1 - 4.8) before your practical session. An example titration and the relevant calculations can be found in Appendix 10.1.

Safety

Chemical Hazard Identification

1 M sodium hydroxide - hazardous. Corrosive, irritant. Avoid eye or skin contact.

potassium hydrogenphthalate - non-hazardous. Low toxicity, irritant.

phenolphthalein - moderate toxicity, irritant. Avoid eye and skin contact.

vinegar - non-hazardous

Risk Assessment and Control

Moderate risk.

Pipettes are easily broken, resulting in dangerous jagged glass edges, especially if the pipette filler is used incorrectly. Read Skill 4.2 & 4.3 for its correct use and practise the techniques described before proceeding.

Waste Disposal

All of the solutions used today can be washed down the sink with water. Solid waste should be thrown in the bin.

Lectura

5. Explique el propósito de este experimento y las habilidades que se adquirirán con esta práctica.
6. Busque en la **Introducción** tres conectores diferentes de ejemplificación.

7. Repaso de voz pasiva. Ubique en el texto cinco ejemplos y tradúzcalos de acuerdo al contexto.

8. ¿Por qué se necesita un proceso de estandarización?

Post Lectura

9. ¿En nuestra facultad se realizan experimentos similares? Ejemplifique.

10. ¿Es peligroso realizar este tipo de experimentos?

TEXTO 27 - 100 YEARS AFTER TITANIC

Opinions

100 years after the Titanic, we're still not unsinkable

By Henry Petroski April 6, 2012

A couple of weeks ago, two other engineers and I were given a tour of a late-1920s mansion outside Detroit. To reach the second floor, we decided to take the vintage elevator. As it descended toward us, it sounded like a cage being rattled by a large animal. Our guide cheerfully told us not to worry; the suspect elevator had been well-maintained and was “absolutely safe.”

My fellow engineers and I exchanged looks. We know that nothing is absolutely safe.

On the 100th anniversary of the Titanic's sinking, it's worth remembering that safety is and always will be relative. Though lifeboat drills have proliferated since 1912, catastrophes still occur. We can't simply blame the engineers when things go wrong because, no matter how well they plan, things don't always go according to plan.

The Titanic is a great example. The ship's owner, the White Star Line, convinced itself and potential passengers that the new vessel was “unsinkable,” but no good engineer should have agreed. Engineers must weigh safety against other features. The Titanic, like all designs, was a compromise.

A century later, it's easy to second- (or third- or fourth- or fifth-) guess the planning of this enormous ship. Conflicts between those who design large technological systems and those who pay for them are resolved by negotiation, a process that doesn't guarantee safety. Thomas Andrews, the Titanic's designer who went down with the ship, wanted bigger bulkheads — watertight walls that separate parts of a ship below decks — and more lifeboats, but White Star wouldn't provide them. The Titanic did have the then-legally required number of lifeboats, but Andrews knew that it was far too few. One row of lifeboats was even removed before sailing to preserve one deck's view.

Meanwhile, the Titanic's bulkheads were not tall enough to compartmentalize the ship under all circumstances. It could have withstood a head-on collision with an iceberg, but not a hole along its side that flooded multiple compartments.

Of course, that's exactly what happened. So much water rushed into the forward compartments that the Titanic's bow dipped. Water cascaded sternward over the tops of the bulkheads and filled compartments further back, sinking the ship. The bulkheads were a fatal flaw.

Had the Titanic not sunk, competing steamship lines may have wanted to one-up White Star by building still larger ships with fewer lifeboats and bulkheads (which restrict passenger movement), resulting in even more dangerous vessels. The sinking provided a wake-up call that fundamentally changed maritime regulation, including the establishment of an International Ice

Patrol. Stronger ships outfitted with enough lifeboats to accommodate passengers and crew became the norm. Overall safety was improved by tragedy.

Today, cruise ships larger than the Titanic have safety and navigation features, such as sonar and radar, that were unavailable to the Titanic's designers. Yet the Costa Concordia, the vessel operated by a subsidiary of Carnival Corp. that ran aground in January off the coast of Italy, had obvious vulnerabilities that modern technology couldn't eliminate. The hull of a vessel so massive could still be ripped apart by a collision with a jagged underwater rock, for example. Everything, even a steel hull, has its breaking point.

With advanced navigation devices giving a captain and his crew constant information about impending obstacles, huge rocks should be easily avoidable. However, on the Concordia, it appears that captain Francesco Schettino could have been emboldened by the very safety features that were supposed to protect his passengers.

"He drives a ship like a Ferrari," one crew member said.

Schettino sailed a more dangerous course than was advisable, exposing his ship to greater-than-expected risks. But state-of-the-art ships the size of the Concordia had navigated shallow waters before. Why couldn't he?

Unfortunately, the Concordia could not steer clear of submerged rocks in time. Its hull was torn open, and the vessel listed badly to starboard. Schettino's decision to drop anchor only made the problem worse, as did a 45-minute delay in deploying lifeboats.

We call the fates of the Titanic and the Concordia — as well as those of the space shuttles Challenger and Columbia — "accidents." Foreseeing such undesirable events is what engineers are expected to do. However, design trade-offs leave technological systems open to failings once predicted, but later forgotten.

Companies selling a product play down its vulnerability and emphasize its robustness. But only after technology leaves the dock is it really tested. For human operators in control of a supposedly infallible system, complacency and overconfidence can take over, and caution may be thrown to the wind. Schettino allegedly wasn't wearing his glasses before the Concordia went down and couldn't read the radar clearly. No ship can be saved from a carefree captain.

Outside Detroit, we three engineers had a rough ride to the mansion's second floor. The elevator started with a jerk and stopped with a greater one. At the tour's conclusion, we rode the elevator back down, this time paying little attention to the racket. After just one trip, we were accustomed to the machine's idiosyncrasies; we ignored the elevator's perceived weaknesses and just asked it to take us where we wanted to go.

How quickly we forget lessons learned.

outlook@washpost.com

Henry Petroski is the Aleksandar S. Vesic professor of civil engineering and a professor of history at Duke University. He is the author of "To Forgive Design: Understanding Failure."

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Resuelva las siguientes consignas

Pre-lectura

- 1) Describa el contexto de producción de este texto. (autor, fuente, fecha, tipo textual, genero textual)

Lectura

- 2) ¿Cuál es la tesis del autor?
- 3) ¿Qué evidencias cita para sustentar su argumento?
- 4) ¿Qué quiere decir el autor con la frase *The Titanic, like all designs, was a compromise*? ¿Qué errores se cometieron en el diseño del Titanic?
- 5) ¿Qué tipo de relación lógica indican los conectores **THOUGH** (Párrafo 3 Línea 1) y **HOWEVER** (L 42)? Mencione las dos ideas que vinculan cada uno.
- 6) ¿Cuál es la conclusión del autor?

Post- lectura

- 7) ¿Está usted de acuerdo con la opinión del autor? Justifique su postura.

TEXTO 28 - CV





EXPERIENCE

Software Engineer

March 2016 - Present

Alden Software Engineering Inc.

The GUI interface provided functions to add, clone and remove profiles from the configuration file at runtime time.

Software Engineer

June 2015 - February 2016

Accu Software

Played key role in designing error analysis architecture for high-availability and supportability.



EDUCATION

Master of Computer Science

May 2013 - May 2015

Bernad University Bernad Street, NYC

Bachelor of Computer Science

April 2009 - April 2013

Bernad University Bernad Street, NYC

High School

March 2005 - March 2009

East High School madson, Wisconsin



SKILLS AND ABILITIES

The following are my best skills which I believe are applicable to the position I have applied for:

1. Programming
2. Logical and Structured Thinking
3. Coding in Multiple Languages



PERSONAL SKILLS

1. Ability to work by myself without the need for constant supervision by employees.
2. Able to work harmoniously with other members of the team.
3. Possesses great analytical and problem solving skills.



ALEX T. SMITH

SOFTWARE ENGINEER



CAREER

A software engineering graduate who is seeking to find the opportunity to work in a fun and challenging working environment that will encourage him to improve and learn new and necessary skills as well as be motivated by the company to do his best for the sake of helping himself and the company advancement in the software engineerig industry.



CONTACT

8100 234 5678

gabloom@email.com

12 M. North Street, Florida



INTEREST





JERICK JERICKSON

NETWORK ENGINEER

Address: 18 Elm Street, New York

Contact Number: 8100 234 5678

Email Address: jerickson@email.com

PROFESSIONAL SUMMARY

Network Engineer with more than 3 years of increasing responsibilities in technical leadership, computer technology, and business-critical management. Expertise in networking concepts, TCP/IP protocol, and network security, knowledge of firewall setup and administration, VPN, bandwidth management, and quality of service. Proven diagnostic abilities with attention to detail and ability to work effectively in a fast-paced environment.

EDUCATIONAL BACKGROUND

MASTERS IN ELECTRICAL ENGINEER

University of California
2012 - 2014
Berkeley, CA, US

BACHELOR OF SCIENCE

University of California
2008 - 2012
Berkeley, CA, US

HIGH SCHOOL

2004 - 2008
Clement High School
Maxwell Rd, Autryville, NC

CAREER OBJECTIVE

To become a reliable network engineer who will be able to assist the company or organization towards achieving its mission & goals.

EMPLOYMENT EXPERIENCE

ARUBA NETWORKS

Networkin Engineer

2016 - Present

Providing daily operational support and system administration for core network infrastructure. Writing up accurate technical documentation.

AGC NETWORKS

Network Engineer

2014 - 2016

Assisting with project scope development. Involved in developing Networking standards, procedures and documentation. Monitoring network performance.

TECHNICAL SKILLS

Critical thinking	Problem solving skills
High technical	Mathematics skills
Active learning	Innovative thinking
Organizational skills	Troubleshooting

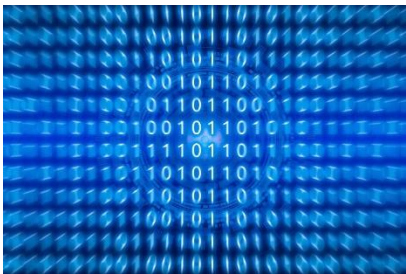
PERSONAL SKILLS

1. Good at finding fast and effective solutions to various Problems
2. Very organized and disciplined.
3. Respectful, well-mannered, and punctual.
4. Willing to learn and is flexible in any source of electrical matters.



A Simple Explanation of 'The Internet of Things'

May 13, 2014



The "Internet of things" (IoT) is becoming an increasingly growing topic of conversation both in the workplace and outside of it. It's a concept that not only has the potential to impact how we live but also how we work. But what exactly is the "Internet of things" and what impact is it going to have on you, if any? There are a lot of complexities around the "Internet of things" but I want to stick to the basics. Lots of technical and policy-related conversations

are being had but many people are still just trying to grasp the foundation of what the heck these conversations are about.

Let's start with understanding a few things.

Broadband Internet is becoming more widely available, the cost of connecting is decreasing, more devices are being created with Wi-Fi capabilities and sensors built into them, technology costs are going down, and smartphone penetration is sky-rocketing. All of these things are creating a "perfect storm" for the IoT.

So what is the Internet of Things?

Simply put, this is the concept of basically connecting any device with an on and off switch to the Internet (and/or to each other). This includes everything from cellphones, coffee makers, washing machines, headphones, lamps, wearable devices and almost anything else you can think of. This also applies to components of machines, for example a jet engine of an airplane or the drill of an oil rig. As I mentioned, if it has an on and off switch then chances are it can be a part of the IoT. The analyst firm [Gartner](#) says that by 2020 there will be over 26 billion connected devices... That's a lot of connections (some even estimate this number to be much higher, over 100 billion). The IoT is a giant network of connected "things" (which also includes people). The relationship will be between people-people, people-things, and things-things.

How does this impact you?

The new rule for the future is going to be, "Anything that can be connected, will be connected." But why on earth would you want so many connected devices talking to each other? There are many examples for what this might look like or what the potential value might be. Say for example you are on your way to a meeting; your car could have access to your calendar and already know the best route to take. **If** the traffic were heavy your car might send a text to the other party notifying them that you could be late. What if your alarm

clock wakes up you at 6 a.m. and then notifies your coffee maker to start brewing coffee for you? What if your office equipment knew when it was running low on supplies and automatically re-ordered more? What if the wearable device you used in the workplace could tell you when and where you were most active and productive and shared that information with other devices that you used while working?

On a broader scale, the IoT can be applied to things like transportation networks: "smart cities" which can help us reduce waste and improve efficiency for things such as energy use; this helping us understand and improve how we work and live. Take a look at the visual below to see what something like that can look like.

The reality is that the IoT allows for virtually endless opportunities and connections to take place, many of which we can't even think of or fully understand the impact of today. It's not hard to see how and why the IoT is such a hot topic today; it certainly opens the door to a lot of opportunities but also to many challenges. Security is a big issue that is oftentimes brought up. With billions of devices being connected together, what can people do to make sure that their information stays secure? Will someone be able to hack into your toaster and thereby get access to your entire network? The IoT also opens up companies all over the world to more security threats. Then we have the issue of privacy and data sharing. This is a hot-button topic even today, so one can only imagine how the conversation and concerns will escalate when we are talking about many billions of devices being connected. Another issue that many companies specifically are going to be faced with is around the massive amounts of data that all of these devices are going to produce. Companies need to figure out a way to store, track, analyze and make sense of the vast amounts of data that will be generated.

So what now?

Conversations about the IoT are (and have been for several years) taking place all over the world as we seek to understand how this will impact **our** lives. We are also trying to understand what the many opportunities and challenges are going to be as more and more devices start to join the IoT. For now, the best thing that we can do is educate ourselves about what the IoT is and the potential impacts that can be seen on how we work and live.

Resuelva las siguientes consignas

- 1- ¿Qué factores se mencionan como facilitadores para la evolución de la Internet de las Cosas?
- 2- Defina y describa de manera completa qué es IoT (Internet of Things). Incluya toda la información relevante. ¿Qué área se menciona para su aplicación? Explique.
- 3- ¿Cuáles son los desafíos que presenta la IoT? Menciónelos y explique por qué son un desafío.
- 4- ¿Qué relación lógica expresa "if" (recuadrado en el 1er párrafo de *"How does this impact you?"*)

¿Qué ideas conecta?

5- ¿A qué hace referencia *our*? (recuadrado en el último párrafo)



Five Biggest Engineering Disasters

The Engineer posted on February 11, 2013 |

As engineers continue to push the boundaries of scientific invention and exploration, along with great achievements unfortunately come terrible failures too. Fortunately, from mistakes lessons are learned but sometimes only once a high price has been paid.

Among the greatest tales of engineering disasters are nuclear reaction explosions, the sinking of great vessels and explosions of space searching rockets. Five of the biggest are the subject of this article and it has been debatable which to include. There were numerous to choose from such as the **World Trade Center**; the Deepwater Horizon Oil Spill; **Hyatt Regency Hotel Walkway Collapse**; the 1970s DC-10 Disasters; **Apollo 1**, and the failure of the New Orleans Levee System during Hurricane Katrina.

While all these disasters remind us of the scale of human endeavour, there is massive potential for failure in the more ordinary. Even bearings, that every day yet essential engineering component of moving systems cannot be overlooked – an overheated one caused an engine failure which caused an engine fire, which led to cabin decompression and finally to the crash of an aircraft in Poland in 1987 – resulting in 193 fatalities.

Engineering is a precise science. Attention to detail and the highest safety standards must be adhered to at all times. Perhaps the most important lesson for us all is not to consign these events to history but to make sure they continue to be recognised by tightened regulations. These lessons also serve as a reminder for people to speak up when necessary to avoid these mistakes being made again.

Chernobyl

In the early hours of 26 April 1986, a structurally unsound reactor at Chernobyl Nuclear Power Plant near Pripyat in Ukraine in the Soviet Union exploded. It caused the release of massive quantities of radioactive particles into the atmosphere which spread over much of Western USSR and Europe. 350,400 people were evacuated and resettled from the most severely contaminated areas of Belarus, Russia, and Ukraine.

The disaster began during a systems test when there was a power surge followed by an emergency shutdown and an exponentially larger spike in power output. The reactor vessel ruptured, there was a series of steam explosions, and the reactor's graphite moderator was exposed to the air, causing it to ignite. The resulting fire sent a plume of highly radioactive fallout into the atmosphere over an extensive geographical area.

In terms of human deaths, the Chernobyl nuclear accident was one of the worst energy accidents in human history. Figures vary widely but between 31 and 64 deaths have been directly attributed to the incident, with estimates of up to 4,000 deaths among those exposed to the highest levels of radiation. Among those living in the broader geographical areas, the numbers of those affected range from 30,000 to nearly a million premature cancer deaths.





The Bhopal Disaster

In 1984, a toxic gas release at a Union Carbide pesticide plant in Bhopal, India resulted in 2,259 immediate deaths and some 11,000 deaths following the disaster.

Over 42 tons of Methyl isocyanate - a highly toxic and irritating material used in the making of pesticides - became contaminated with water, causing an exothermic reaction, which increased the temperature inside the tank to over 200°C, far beyond its capacity. Automated emergency release systems kicked in, venting the extra pressure and a large volume

of gasses, which escaped and began to spread. If that gas had been lighter than air, it might have dispersed with minimal harm. Unfortunately, being heavier than air, it crawled for miles, seeping into the nearby city of Bhopal. A government affidavit in 2006 stated the leak caused 558,125 injuries including 38,478 "temporary partial" and approximately 3,900 "severely and permanently disabling" injuries.

It has now been 25 years since the horrific disaster, yet still the land around Bhopal remains blighted and toxic to humans and animals alike. Today, the 390 tons of toxic chemicals continue to pollute the groundwater in the region. The disaster serves as a reminder for the way the industry approaches process safety management (PSM). The right decisions need to be made daily to ensure the safety of your process – and this in turn helps ensure the meaning of the phrase “recognised and generally accepted good engineering practices.”

Titanic

The sinking of the Titanic is history's most epic sea disaster and the topic of endless books, feature films and documentaries.

In 1912, the world's largest ship, on its maiden voyage carrying some of the world's richest people had a freak accident with a catastrophic loss of life. Four days into her journey, on the night of 14th April the **Titanic** struck an iceberg in the North Atlantic and was so badly damaged that she survived for less than three hours before she sank. Two thirds of Titanic's passengers and crew were lost because there were not enough lifeboats to rescue everyone on board. Over 1500 people drowned.



It is believed that safety took second place to aesthetics in the ship's design. The original design included two rows of lifeboats on the deck, but one row was removed allowing more space and a better view for passengers with first-class berths. This was not against the Board of Trade regulations which covered only ships up to 13,000t and not the 46,000t Titanic.

Research has found it likely that the iceberg buckled the plates and popped out substandard rivets along a length of the hull, allowing water into at least five of the watertight compartments. As the boat went down by the head, water flowed over the transverse bulkheads, which were barely above the waterline, into other compartments. It has been said that the height of the bulkheads was reduced to avoid spoiling the first-class public rooms.

In the aftermath of the disaster, the height of transverse bulkheads was increased and double hulls reaching further up the sides of ships became common. There were new regulations regarding safety, increasing the number of lifeboats and allowing easy access to them for all passengers. There were also changes regarding the use of radio at sea.



The Space Shuttle Challenger

Just 73 seconds after its launch, on 28 January 1986, space shuttle Challenger broke apart, killing its seven crew members. The subsequent Rogers Commission found the cause of the accident was the failure of both primary and secondary O-rings on the right solid rocket booster, allowing hot gas and flame to escape, which then came into contact with the booster attachment and external tank, resulting in structural failure. The problems with the O-rings had been known about for nine years but had been ignored, partly because safety was deemed ensured with the presence of the

second ring. However, as was later made clear, the second ring was there for unforeseen failure, not a failure that had been considered. Engineers' warnings that low temperatures would exacerbate the problem were also ignored by NASA managers because of pressure to keep to the launch timetable. Now widely used as a case study for trainee engineers, this disaster has been used to teach many lessons: primarily that the advice of engineers should be considered carefully by management; and that the ethics of whistle-blowing and group decision-making should be introduced. Afterwards, there was a total redesign of the solid rocket boosters, in which three O-rings were incorporated, watched over by an independent oversight group as stipulated by the commission.

In summing up the disaster, Richard Feynman, a member of the Rogers Commission, made a telling point to the effect that "for a successful technology, reality must take precedence over public relations, for nature cannot be fooled".



Apollo 13

The disaster about the crippled flight to the moon in 1970 that gave the film industry two of the most iconic film lines: "Houston, we have a problem" and "failure is not an option".

Apollo 13 was the seventh manned mission in the American Apollo space program and the third intended to land on the Moon. The craft was launched on April 11, 1970, at 13:13 CST from the Kennedy Space Center, Florida, but the lunar landing was aborted after an oxygen tank exploded two days later, crippling the service module upon which the Command Module depended.

Despite great hardship caused by limited power, loss of cabin heat, shortage of potable water, and the critical need to jury-rig the carbon dioxide removal system, the crew returned safely to Earth on April 17.

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Resuelva las siguientes consignas

Primera aproximación al texto:

1. ¿Qué tipo de texto es? ¿Dónde encontraría este tipo de texto?
2. Complete el siguiente cuadro:

Título	Fuente	Fecha de publicación

3. Lea los subtítulos y analice con sus compañeros qué conocimientos tiene al respecto.

Lesson 21: Free Body Diagrams

Free Body Diagrams

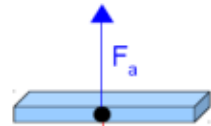
We will need a way to organize forces that are acting on a particular object. The easiest way to do this is by using a **free body diagram**.

- A free body diagram is just a *simple* sketch of the object showing all the forces that are acting on it.
- To draw a proper free body diagram, you must follow these steps:
 1. Draw a quick **sketch** of the object. Often a simple box will do.
 2. Place a **dot** in the centre of the object. We basically treat this as the spot that all the forces are thought to act upon.
 3. For every force acting on *that* object (we don't care about forces acting on any other objects), draw a **vector** that shows the size and direction of the force. Each vector must start from the dot and point outwards.
 4. **Label** each vector based on the type of force it is. *Do not include numbers or calculations!*

Example 1: Sketch a free body diagram for a book being held up by a person.

We draw a quick sketch of the book, and then put a dot in the centre.

Next, we identify that there are two forces acting on this book; the force of gravity pulling it down, and the applied force of a person pushing it up. Since the book is just being held up (not accelerating up or down), we can assume that the two forces are equal, so we will draw the vectors the same size.



Drawing 1: Free body diagram of forces on a book.

Free body diagrams must show a wide variety of forces acting on an object.

- These are the common forces acting on objects that you need to memorize:

F_g = force due to gravity
 F_a = applied force
 F_f = force due to friction
 F_T = force of tension
 F_N = normal force
 F_{NET} = net force

Warning!

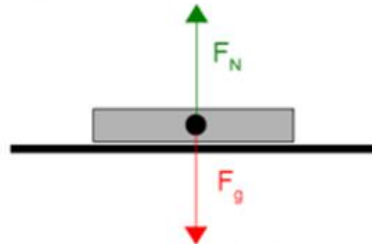
It is critical that you label the forces on your diagrams using exactly the symbols shown here. The use of capitals and lower case letters does matter!

Each of the forces that have not been mentioned yet will be discussed in detail as they are introduced in the following examples.

Example 2: Sketch a free body diagram of a laptop sitting on a table.

- We know that there will be a force due to gravity (F_g) pulling the laptop down, but if that was the only force it should be dropping down towards the ground.
- There must be a force acting against the force of gravity that is holding the laptop up. This is happening because the table top is strong enough to hold up the laptop.
 - We call this upwards force the **normal force** (F_N). *A normal force is exerted upwards by a surface (like a table or a floor) and is **perpendicular** to the surface.*

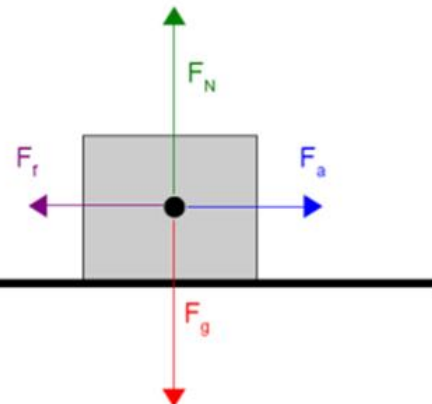
- As long as the object is not breaking through the table top or flying up into the air, the force due to gravity and the normal force must be equal. That way they just cancel out and the object stays where it is.



Drawing 2: Free body diagram for example 2.

Example 3: You are trying to push a heavy box across the floor. It's causing you some trouble because the floor is not very smooth. **Sketch** a free body diagram of the box.

- We still have force due to gravity and normal force, since it is still an object on a surface.
- You are trying to push it sideways, so that will be an applied force that you are exerting.
- It's tough to push because of friction between the box and the floor, so we'll also need to draw a force due to friction.
 - **Force due to friction (F_f)** always opposes the motion of an object. It is parallel to the surface the object is on.
- Notice the normal force and force due to gravity are still equal. The applied force and force due to friction are also equal to each other. This would mean that the object is moving at a constant velocity (more on this in later sections).



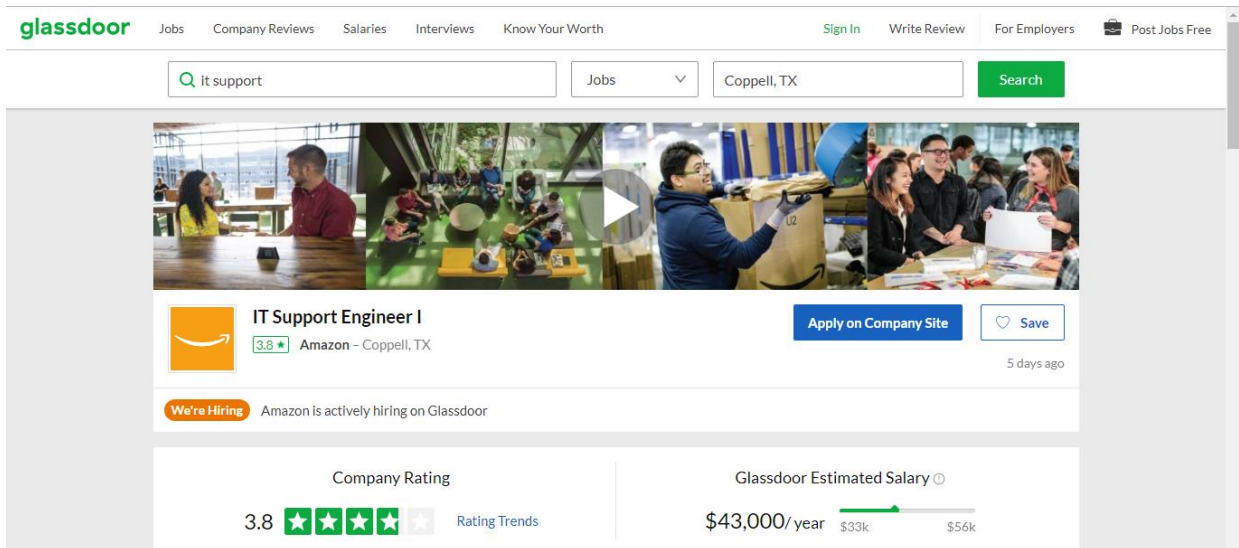
Drawing 3: Free body diagram for example 3.

Responda las siguientes consignas:

- 1) ¿Qué es un diagrama de cuerpo libre y para qué se utiliza?
- 2) ¿Qué pasos se deben seguir para dibujar un diagrama de cuerpo libre?
- 3) Mencione las fuerzas que actúan sobre los objetos y qué debemos tener en cuenta al escribir sus símbolos.
- 4) Explique el ejemplo que describe cómo dibujar un diagrama de cuerpo libre de una computadora portátil apoyada sobre una mesa.

- 5) Explique las fuerzas involucradas cuando se quiere empujar una caja pesada sobre el piso.

TEXT0 32 - JOB AD



The screenshot shows the Glassdoor website interface. At the top, there's a navigation bar with links for Jobs, Company Reviews, Salaries, Interviews, and Know Your Worth. On the right, there are links for Sign In, Write Review, For Employers, and Post Jobs Free. Below the navigation bar is a search bar with the text "it support" entered. To the right of the search bar are dropdown menus for "Jobs" and "Coppell, TX", and a green "Search" button. Below the search bar is a video player showing a collage of images related to the job. Below the video player is the job title "IT Support Engineer I" and the company name "Amazon - Coppell, TX". To the right of the job title is a blue button "Apply on Company Site" and a white button "Save". Below the job title is a "We're Hiring" badge and the text "Amazon is actively hiring on Glassdoor". Below the job title is a "Company Rating" section showing a 3.8 star rating and a "Glassdoor Estimated Salary" section showing a salary range of \$43,000/year to \$56k.

glassdoor Jobs Company Reviews Salaries Interviews Know Your Worth Sign In Write Review For Employers Post Jobs Free

it support Jobs Coppell, TX Search

IT Support Engineer I Amazon - Coppell, TX Apply on Company Site Save 5 days ago

We're Hiring Amazon is actively hiring on Glassdoor

Company Rating 3.8 Rating Trends

Glassdoor Estimated Salary \$43,000/year \$33k \$56k

Job Description

At Amazon, we strive to be Earth's most customer-centric company where people can find and discover anything they want to buy online. We hire the world's brightest minds, offering them an environment in which they can relentlessly improve the experience for customers. Innovation and creativity are built into the DNA of the company and are encouraged at all levels of employment. Every day we solve complex technical and business problems with ingenuity and simplicity. We're making history and the good news is we've only just begun.

Amazon, one of the top 100 companies in the United States, has an immediate opening for an IT Support Engineer in our Fulfillment Center. Fulfillment Centers throughout the U.S. handle millions of items from tens of thousands of vendors and ships them across the globe each day. The successful IT Support Engineer will be responsible for utilizing his/her skills to provide assistance to the operations team in resolving technical problems within the Fulfillment Center that cover a multitude of technical disciplines. This position is the second level of support for the operations of the Fulfillment Center and serves as a resource to Technical Support Technicians while also acting as a representative of the IT department to internal customers. Amazon seeks individuals who are willing to Work Hard, Have Fun, and join us in making Worldwide History.

The IT Support Engineer must have an excellent understanding of computers (both hardware and software) and a demonstrated willingness to learn and apply new technology. They must possess superior technical aptitude, written and oral communications skills and the ability to deal effectively with people in a wide variety of situations. Additionally, they must possess strong analytical skills with demonstrated problem solving ability. Previous experience exercising high levels of initiative, judgment, and diplomacy required. Proven ability to operate in extreme high pressure situations, successfully handle multiple priorities and possess a high level of discipline while remaining flexible. They must possess superior people skills dealing with all levels of a team environment up to and including presentations and explanation of projects to upper level management. Regular activities include network engineering and troubleshooting, project management, mentorship of Technical Support Technicians, data cabling, systems administration in a variety of software and hardware

environments, telecom/PBX administration, root cause analysis in problem solving, and assistance in managing the daily activities of the department. A willingness to relocate as career advancement opportunities throughout the company become available is strongly desired.

Basic Qualifications

- High School diploma or GED equivalent
- Expertise in one or more specialized areas of IT such as Microsoft Administration, Linux Administration, or Cisco IOS (CLI)
- Advanced Troubleshooting skills in a multi-user high pressure environment
- Understanding of networking concepts such as DNS, DHCP, Email, HTTP, SSL, OSI Model, and TCP/IP protocols and applications
- Experience with Data Cabling / Computer Facilities maintenance
- Proven skills in PC repair, troubleshooting, deployment, and liquidation
- Ability to stand/walk for 10-12 hours per day
- Willingness to work flexible shifts and scheduling including nights, weekends, and holidays

Preferred Qualifications

- Bachelor's degree in Computer science or IT related field.
- Microsoft MCSE, MCITP Systems Administrator (Active Directory)
- Cisco CCNA certification
- Advanced Linux systems administration skills
- Strong troubleshooting skills of very complex systems
- Ability to explain complex IT concepts in simple terms
- Ability to manage high priority projects
- Proven ability to work successfully with limited supervision
- Excellent written and verbal communication skills. Comfortable communicating in person and via phone/radio with peers, management, contractors and vendors.
- Ability to travel up to 25%

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<https://www.glassdoor.com>



Working conditions of workers of different ages

Introduction

Demographic ageing implies a greater participation by older people in the labour market. To enable this, it is important to examine the factors that can facilitate or hinder extended working lives. Making work more sustainable over the life course requires working conditions that support individuals in remaining in work until the statutory retirement age.

Many factors influence sustainable work: individual characteristics, work-related elements, social norms and the institutional context. This report focuses on the role of working conditions in shaping sustainable work over the life course, using data from Eurofound's European Working Conditions Survey (EWCS 2015). The report investigates working conditions for different age groups, and links these conditions to three sustainable work outcomes: work-life balance, health and well-being, and career prospects. It also investigates differences in working conditions on the basis of gender, occupation and country.

Policy context

In recent years, many Member States have raised the retirement age and provided financial incentives to work at older ages – in a few cases, seeking to improve working conditions. At EU level, the European Pillar of Social Rights aims to foster convergence towards better working and living conditions. Its principles include efforts to make work sustainable, such as: training and lifelong learning; flexible working arrangements for better work-life balance; and the right to a working environment adapted to workers' needs. Both the Pillar and the Europe 2020 strategy seek to increase employment rates across the board, suggesting that the focus on improving working conditions must extend to include all age groups. And in the recently concluded autonomous agreement on active ageing, the European social partners commit to facilitating older workers in actively participating in the labour market until retirement age.

Key findings

Poor working conditions have a negative impact on sustainable work outcomes for all employees, regardless of age. Employees who are exposed to physical risks and quantitative demands (working at high speed and to tight deadlines) are more likely to experience worse health and poorer work-life balance. They are also more likely to state that they will not be able to continue work until age 60. Intention to depart the workforce earlier is also significantly associated with poor-quality management and experiencing adverse social behaviour.

Some aspects of working conditions remain stable or deteriorate until the age of 55 and then improve. Workers aged 55 and above report less exposure to physical risks, shorter weekly working hours and greater working time autonomy; older employees also report better work-life balance. However, older employees also participate less in training, and have more limited career prospects.

Workers aged 45–54 years report fewer quantitative demands but their level of exposure to physical risks is nearly as high as that of younger workers.

A critical issue for employees aged 35–44 years is work-life balance, since they tend to work longer hours and have more care responsibilities.

Results are mixed for younger employees (aged 35 and under) regarding the social environment at work. They are more likely to have social support and positive encouragement from colleagues and their boss, but also experience more adverse social behaviour – particularly women. They are also the most likely to work on temporary contracts, which may contribute to the greater job insecurity they report.

An analysis of the working conditions of workers of different ages also has to take into account differences between occupations. For all ages, a lower occupational level is associated with poorer health and well-being, and poorer career prospects. Especially for low- and mid-level

occupations, working conditions are consistently poor over the life course. The impact of poor working conditions from a young age is likely to accumulate, resulting in poor outcomes at an older age.

Unsurprisingly, working conditions vary by country and by age group across countries: in Hungary and Greece, work-life balance among older employees is much poorer than elsewhere. And for workers within the same occupation, country differences in working conditions outcomes are also evident.

Sustainable work outcomes affect the expected duration of working life – not necessarily in expected ways. Belgium, for example, with generally a good picture of sustainable work, has a shorter duration of working life. Estonia, in contrast, with poorer results for sustainable work outcomes, has a longer duration of working life and higher employment rates at older ages. These differences could be due to different institutional arrangements that facilitate or hinder early retirement.

Policy pointers

Importance of working conditions for workers aged 55+: It is precisely at the age of 55 that participation in work strongly declines. Conditions of work such as good physical environment, lower quantitative demands and working time autonomy increase the likelihood of workers continuing working when they are aged 55 and over.

Recognise role of national institutional context: Institutional and social contexts play an important role in determining duration of working life. Extending working life hence depends on measures in the areas of pension systems, health, education and care.

Workplace social partners have a particular role:

Differences between countries in relation to working conditions have to be considered, for example, for the implementation of the Social Partners' Framework Agreement on Active Ageing at national level. The role that employers' and workers' representatives can play at national, sectoral and workplace level is essential.

Broaden definitions of 'arduous jobs': Occupations where work over the life course has a negative effect on sustainable work outcomes require particular attention. The physical context, work organisation and social environment (including psychosocial risks) need to be considered. In line with this, in countries where arduous jobs are defined and given special treatment, psychosocial risks should be factored in.

Prioritise training for older employees: One of the challenges that demographic change poses is that of maintaining and updating the skills of the workforce. To increase the share of older employees involved in learning and on-the-job training, implementing the first principle of the European Pillar of Social Rights (on-the-job training for employees over the age of 45) should be a priority.

Monitor trends in world of work: Changes in the world of work are likely to have implications on working conditions across all ages and could impact on sustainable work over the life course. For example, the use of temporary contracts (currently limited largely to younger employees) may spread to other age groups with potential consequences. At the same time, technological developments associated with the digitalisation of work may offer opportunities for older workers' participation. Research and policy work need to monitor these developments.

Further research required on role of motivation: Finally, motivation has been identified in the literature review as one of the key factors contributing to remaining at work until older age. This aspect has to be further investigated.

Resuelva las siguientes consignas

Prelectura

1. Título:
2. Fuente de publicación: ¿A qué se refiere dicha fuente?
3. Este artículo es un resumen ejecutivo. Busque con su compañero en internet la siguiente información:
 - Definición de resumen ejecutivo
 - Objetivos

- Quién redacta este tipo de artículos y a quién está dirigido
- Partes que no deben faltar en un resumen ejecutivo

Lectura

4. ¿Cumple este resumen con todas las partes?
5. Lea la 1ra oración de los dos párrafos introductorios y explique qué más se puede complementar al título.
6. Explique con sus palabras el contexto en el que se desarrolla este resumen.
7. Fundamente por qué el autor hace uso de la frase “A critical issue”. Tenga en cuenta el conector mencionado en la oración.
8. Sintetice las indicaciones sobre políticas que hay que tener en cuenta al respecto. Elija una que le parezca más importante en el caso de Argentina, y explíquela. ¿En qué basa su elección?

Marie Curie Biography

Physicist, Scientist, Scientist (1867–1934)

BIOGRAPHY

Marie Curie was the first woman to win a Nobel Prize, in Physics, and with her later win, in Chemistry, she became the first person to claim Nobel honors twice.



Marie Curie working in her laboratory at the University of Paris in 1925. (AFP/Getty)

Born Maria Skłodowska on November 7, 1867, Marie Curie became the first woman to win a Nobel Prize and the first person—man or woman—to win the award twice. Curie's efforts, with her husband Pierre Curie, led to the discovery of polonium and radium and, after Pierre's death, the further development of X-rays.

Marie Curie discovered radioactivity, and, together with her husband Pierre, the radioactive elements polonium and radium, while working with the mineral pitchblende. Fascinated with the work of Henri Becquerel, a French physicist who discovered that uranium casts off rays weaker than the X-rays found by Wilhelm Conrad Roentgen, Marie Curie took his work a few steps further. Curie conducted her own experiments on uranium rays and discovered that they remained constant, no matter the condition or form of the uranium. The rays, she theorized, came from the element's atomic structure. This revolutionary idea created the field of atomic physics. Curie herself coined the word "radioactivity" to describe the phenomena.

Following Marie's discovery of radioactivity, she continued her research with her husband. Working with the mineral pitchblende, the pair discovered a new radioactive element in 1898. They named the element polonium, after Marie's native country of Poland. They also detected the presence of another radioactive material in the pitchblende and called that radium. In 1902, the Curies announced that they had produced a decigram of pure radium, demonstrating its existence as a unique chemical element.

Marie married French physicist Pierre Curie on July 26, 1895. They were introduced by a colleague of Marie's after she graduated from the University of Sorbonne; Marie had received a commission to perform a study on different types of steel and their magnetic properties and needed a lab to work in. A romance developed between the brilliant pair, and they became a scientific dynamic duo who were completely devoted to one another. At first Marie and Pierre worked on separate projects. But after Marie discovered radioactivity, Pierre put aside his own work to help her with her research.

Marie suffered a tremendous loss in 1906, when Pierre was killed in Paris after accidentally stepping in front of a horse-drawn wagon. Despite her tremendous grief, she took over his teaching post at the Sorbonne, becoming the institution's first female professor.

In 1911, Marie Curie's relationship with her husband's former student, Paul Langevin, became public. Curie was derided in the press for breaking up Langevin's marriage, the negativity in part stemming from rising xenophobia in France. In 1897 Marie and Pierre Curie welcomed a daughter, Irène. The couple had a second daughter, Ève, in 1904. Irène Joliot-Curie followed in her mother's footsteps, winning the Nobel Prize in Chemistry in 1935. Joliot-Curie shared the honor with her husband, Frédéric Joliot, for their work on the synthesis of new radioactive elements.

Both of Marie Curie's parents were teachers, and she was the youngest of five children, following siblings Zosia, Józef, Bronya and Hela. As a child Curie took after her father, Wladyslaw, a math and physics instructor. She had a bright and curious mind and excelled at school. However, tragedy struck early: When she was only 10, Curie lost her mother, Bronislawa, to tuberculosis.

A top student in her secondary school, Curie could not attend the men's-only University of Warsaw. She instead continued her education in Warsaw's "floating university," a set of underground, informal classes held in secret. Both Curie and her sister Bronya dreamed of going abroad to earn an official degree, but they lacked the financial resources to pay for more schooling. Undeterred, Curie worked out a deal with her sister. She would work to support Bronya while she was in school and Bronya would return the favor after she completed her studies. For roughly five years, Curie worked as a tutor and a governess. She used her spare time to study, reading about physics, chemistry and math.

In 1891, Curie finally made her way to Paris and enrolled at the Sorbonne. She threw herself into her studies, but this dedication had a personal cost. With little money, Curie survived on buttered bread and tea, and her health sometimes suffered because of her poor diet. Curie completed her master's degree in physics in 1893 and earned another degree in mathematics the following year.

Marie Curie was the first woman to win a Nobel Prize and the first person—man or woman—to win the prestigious award twice. She remains the only one to be honored for accomplishments in two separate sciences. In 1903, Curie received the Nobel Prize in Physics, along with her husband and Henri Becquerel, for their work on radioactivity. With their win, the Curies developed an international reputation for their scientific efforts, and they used their prize money to continue their research.

In 1911, Curie won her second Nobel Prize, this time in Chemistry, for her discovery of radium and polonium. While she received the prize alone, she shared the honor jointly with her late husband in her acceptance lecture. Around this time, Curie joined with other famous scientists, including Albert Einstein and Max Planck, to attend the first Solvay Congress in Physics and discuss the many groundbreaking discoveries in their field.

When World War I broke out in 1914, Curie devoted her time and resources to helping the cause. She championed the use of portable X-ray machines in the field, and these medical vehicles earned the nickname "Little Curies." After the war, Curie used her celebrity to advance her research. She traveled to the United States twice—in 1921 and in 1929—to raise funds to buy radium and to establish a radium research institute in Warsaw.

Marie Curie made many breakthroughs in her lifetime. Remembered as a leading figure in science and a role model for women, she has received numerous posthumous honors. Several educational and research institutions and medical centers bear the Curie

name, including the Curie Institute and Pierre and Marie Curie University, later renamed UPMC.

In 1995, Marie and Pierre Curie's remains were interred in the Panthéon in Paris, the final resting place of France's greatest minds. Curie became the first and one of only five women to be laid to rest there. In late 2017, the Panthéon hosted an exhibition to honor the 150th birthday of the pioneering scientist.

In 1937, Ève Curie wrote the first of many biographies devoted to her famous mother, *Madame Curie*, which became a feature film a few years later. The story of the Nobel laureate was back on the big screen in 2017 with *Marie Curie: The Courage of Knowledge*, featuring Polish actress Karolina Gruszka. In 2018, it was announced that Amazon Prime Video was developing another biopic of Curie, with British actress Rosamund Pike in the starring role.

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Source: <https://www.biography.com/people/marie-curie-9263538>

Resuelva las siguientes consignas

1. ¿Qué sabe de Marie Curie?
2. Realice una lectura rápida del texto buscando la información que mencionó anteriormente.
3. ¿Dónde ubicaría los siguientes subtítulos?

What Did Marie Curie Discover?

How and When Did Marie Curie Die?

When and Where was she Born?

Nobel Prizes

Family

X-rays and Later Research

Who Was Marie Curie?

Education

Legacy



A) The usual answer to this old riddle is as follows. If it takes three cats three minutes to catch three rats, it must take them one minute to catch one rat. And if it takes them a minute for each rat, then the same three cats would catch 100 rats in 100 minutes.

Unfortunately, it is not quite that simple; such an answer presupposes something that is certainly not stated in the problem. It assumes that all three cats concentrate their attention on the same rat until they catch him in one minute, then turn their combined attention toward another rat. But suppose that instead of doing this, each cat chases a different rat and takes three minutes to catch it. In this case, three cats would still catch three rats in three minutes. It would take them six minutes to catch six rats, nine minutes to catch nine rats, and 99 minutes to catch 99 rats.

A curious difficulty now faces us. How long will it take those same three cats to catch the 100th rat? If it still takes them the full three minutes to run him down, then it will take three cats 102 minutes to catch 100 rats.

To catch 100 rats in 100 minutes - assuming this is how the cats go about their rat catching -we will certainly need more than three cats and less than four cats.

Of course it is possible that when the three cats gang up on a single rat they can corner him in less than three minutes, but there is nothing in the statement of the riddle that tells us exactly how to measure the time for this operation. The only correct answer to the problem, therefore, is this: The question is ambiguous and cannot be answered without more information about how those cats catch rats.

B) Surprisingly, the second offer is much better than the first one. If you accept it, you will earn exactly \$200 more each year than you would on the other basis of payment! The chart below shows your total earnings, on the basis of both offers, for each of the first six years.

<i>Years</i>	<i>Offer A</i>	<i>Offer B</i>
1	\$4,000	\$4,200
2	4,800	5,000
3	5,600	5,800
4	6,400	6,600
5	7,200	7,400
6	8,000	8,200